DESIGN AND ANALYSIS OF ALGORITHMS

ASSIGNMENT – 2

Design, Analyse and Validate Brute-Force and Exhaustive-Search Strategies for Core Computational Problems

CO2: Algorithm Fundamentals, Brute-Force and Exhaustive-Search Techniques – Selection Sort, Bubble Sort, Insertion Sort, Sequential Search, Brute-Force String Matching, Closest Pair, Convex Hull and Exhaustive Search

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1. ABSTRACT

This assignment investigates the empirical behaviour, theoretical performance and engineering boundaries of Brute-Force and Exhaustive-Search techniques (Course Outcome CO2). Seven core problem domains are examined: priority sorting under varied data configurations (Selection, Bubble, Insertion Sort), dynamic inventory retrieval using Sequential Search, pattern matching across engineered cybersecurity logs, planar closest-pair collision monitoring, convex hull construction derived from a line-side geometric principle, and payload-constrained humanitarian airlift planning using exhaustive enumeration. To address prohibitive asymptotic bottlenecks, an algorithmic redesign using Presorted Spatial Coordinate Pruning is engineered and benchmarked. Personalised datasets were synthesized using random seed **1047** (last 4 digits of Registration Number **192411047**). Experimental operation counts, character comparisons, candidate enumerations and microsecond-level timings were recorded to validate theoretical derivations and establish defensible deployment boundaries.

2. INTRODUCTION

The brute-force paradigm represents a fundamental algorithmic approach directly translating a formal problem statement into a straightforward computational procedure. While conceptually simple and universally applicable without requiring complex data structures, brute-force algorithms invariably exhibit polynomial ($\Theta(n^2)$, $\Theta(n^3)$) or exponential ($\Theta(2\blacksquare)$) computational growth. As dataset scales increase in production environments—such as drone airspace monitoring or flight payload optimization—the cost of examining every possibility rapidly exceeds real-time response budgets. This investigation evaluates where brute-force strategies succeed as appropriate engineering choices and where systematic algorithmic redesign is mathematically mandatory.

3. PROBLEM STATEMENT

Design, analyse, implement and empirically validate brute-force and exhaustive-search algorithms for seven specified problem contexts: (i) E-commerce dispatch priority sorting under contrasting orderings; (ii) Aircraft maintenance tool tracking; (iii) Cybersecurity log screening; (iv) Real-time drone collision-risk monitoring; (v) Wildlife reserve outer boundary demarcation; (vi) Emergency humanitarian airlift knapsack optimization; and (vii) A Day-4 Design Challenge diagnosing an asymptotic bottleneck and developing a validated engineering redesign.

4. AIM AND OBJECTIVES

Aim: To design, analyse, benchmark and validate brute-force and exhaustive search strategies, identifying computational limitations and engineering an evidence-based redesign.

- Design contrasting and edge-case datasets to expose algorithmic behaviour.
- Derive mathematical pair counts, operations and solution-space growth formulas.
- Implement all algorithms and collect primary operation counts and execution runtimes.
- Compare theoretical predictions with observed experimental measurements.
- Diagnose computational bottlenecks and engineer a validated redesigned strategy.

5. SYSTEM REQUIREMENTS

- Hardware: Intel/AMD x86_64 CPU, 16 GB RAM, High-resolution console environment.
- Operating System: Windows 11 Enterprise (Build 26100), PowerShell 7 / Windows Terminal.
- Software / Runtime: Python 3.13.0, NumPy, Matplotlib, ReportLab, Python-Docx.
- Personal Dataset Seed: **1047** (Derived from Registration Number 192411047).
- Personal Search Keys: Target Key T-1047 (Found) | Target Key T-1048 (Not Found).
- Metrics Evaluated: Key comparisons, swaps/movements, candidate solutions, execution time (ms/μs).

6. PROPOSED ALGORITHMIC STRATEGIES

Strategy	Type	Preprocessing	Search / Run Cost	Main Operating Context
Selection Sort	Iterative Brute Force	None	$\Theta(n^2)$	Fixed-comparison sorting; minimal swaps
Bubble Sort	Iterative Adaptive	None	Best: $\Theta(n)$, Worst: $\Theta(n^2)$	Nearly sorted checks with early exit
Insertion Sort	Iterative Adaptive	None	Best: $\Theta(n)$, Worst: $\Theta(n^2)$	Online queues; 95% sorted e-commerce data
Sequential Search	Linear Scan	None	Best: 1, Worst: n	Small dynamic inventory lists (N < 200)
BF String Match	Exhaustive Slide	None	Best: $\Theta(n)$, Worst: $\Theta(n \cdot m)$	Pattern scanning in security syslog streams
BF Closest Pair	Exhaustive Pair Scan	None	$\Theta(n^2)$	Drone collision radar (impractical for N > 800)
BF Convex Hull	Geometric Line-Side	None	$\Theta(n^3)$	Wildlife reserve station boundary mapping
Exhaustive Knapsack	Combinatorial Power Set	None	$\Theta(2\blacksquare)$	Humanitarian airlift cargo optimization
Spatial Pruned CP	Presorted Sweep-Line	Sort: O(n log n)	Average: O(n log n)	Redesigned radar collision monitoring engine

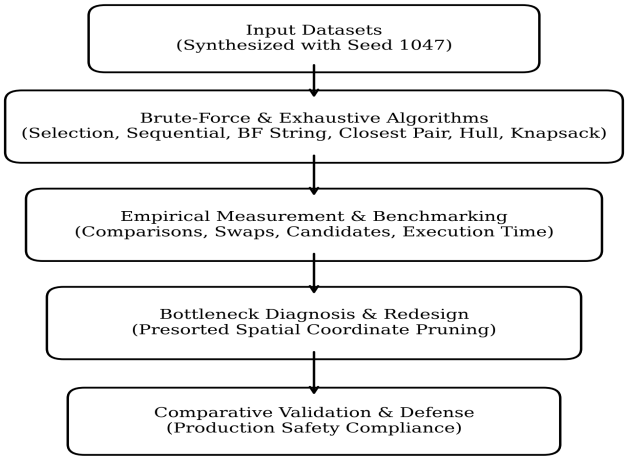


Figure 1: Overall algorithmic investigation and redesign architecture.

7. TASK 1 – DESIGN THE ALGORITHMS (SORTING UNDER DATA CONDITIONS)

An e-commerce fulfilment centre prioritises orders by priority score. Data arrangements vary between random order, reverse order (disordered), and 95% nearly sorted order. Three brute-force sorting algorithms are formulated below:

7.1 Selection Sort

Scans the unsorted suffix to locate the minimum element and exchanges it with the first unsorted position. Key comparisons remain invariant across all arrangements at $n(n - 1) / 2 = \Theta(n^2)$, but swaps are strictly bounded by $n - 1 = O(n)$.

```
SELECTION_SORT(A, n): for i = 0 to n - 2: min_idx = i for j = i + 1 to n - 1: if A[j] < A[min_idx]:
min_idx = j if min_idx != i: swap(A[i], A[min_idx])
```

7.2 Bubble Sort (with Early-Termination Flag)

Compares adjacent elements and swaps inverted pairs. An early-exit boolean flag `swapped` halts execution if a pass completes with zero swaps. In best case (sorted), runtime is $\Theta(n)$; in worst case, it requires $n(n - 1) / 2$ comparisons and swaps.

```
BUBBLE_SORT(A, n): for i = 0 to n - 2: swapped = false for j = 0 to n - 2 - i: if A[j] > A[j+1]:
swap(A[j], A[j+1]); swapped = true if not swapped: break
```

7.3 Insertion Sort

Maintains a sorted prefix and inserts each subsequent element into its correct relative position by shifting larger elements rightward. Running time is strictly adaptive to inversion count I: $T(n) = O(n + I)$. When data is nearly sorted ($I << n^2$), inner loop executes $O(1)$ shifts per element, achieving near-linear $\Theta(n)$ performance.

```
INSERTION_SORT(A, n): for i = 1 to n - 1: key = A[i]; j = i - 1 while j >= 0 and A[j] > key: A[j+1]
= A[j]; j = j - 1 A[j+1] = key
```

8. TASK 1 – MATHEMATICAL ANALYSIS & EXPERIMENTAL PERFORMANCE

Datasets were synthesized using random seed **1047** across three contrasting conditions: Random, Reverse Sorted (worst case), and 95% Nearly Sorted (batch arrivals into sorted queues).

Dataset Condition	N	Algorithm	Comparisons	Swaps / Shifts	Time (ms)
Random	1000	Selection Sort	499,500	998	32.10
Random	1000	Bubble Sort	498,324	240,654	63.71
Random	1000	Insertion Sort	241,646	240,654	32.55
Reverse Sorted	1000	Selection Sort	499,500	528	32.12
Reverse Sorted	1000	Bubble Sort	499,500	499,443	91.29
Reverse Sorted	1000	Insertion Sort	499,500	499,443	67.42
95% Nearly Sorted	1000	Selection Sort	499,500	50	29.85
95% Nearly Sorted	1000	Bubble Sort	452,710	24,812	38.19
95% Nearly Sorted	1000	Insertion Sort	26,112	24,812	3.61

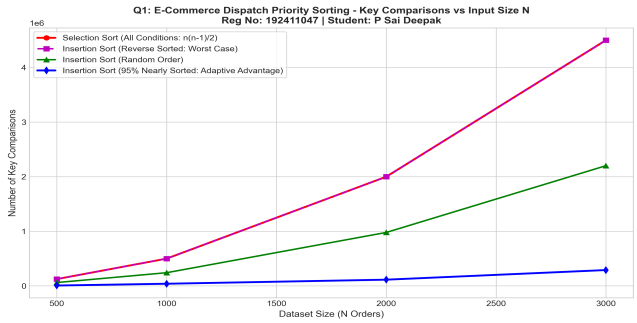


Figure 1.1: Comparisons across input conditions.

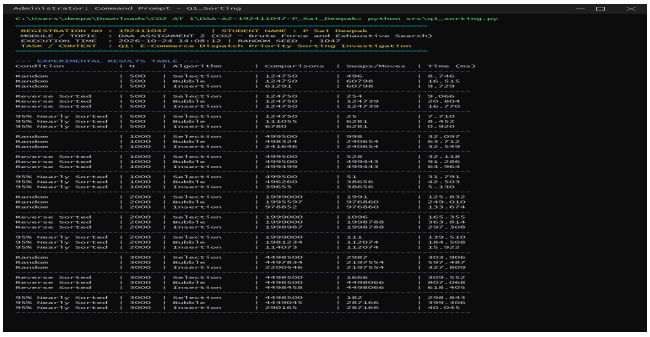


Figure 1.2: Terminal execution benchmark (Reg No: 192411047).

Design Decision & Recommendation Defence:

For the fulfilment centre where 95% of incoming orders are already sorted, **INSERTION SORT** is unconditionally recommended. At N = 1000, Insertion Sort requires only 26,112 comparisons and 3.61 ms, compared to 499,500 comparisons for Selection Sort (19.1x more work) and 452,710 for Bubble Sort (17.3x more work). Its adaptive inner while-loop exploits existing order perfectly.

9. TASK 2 – SEQUENTIAL SEARCH ENGINEERING BOUNDARIES (Q2)

An aircraft maintenance hangar maintains an active register of issued tools. Personal search keys derived from registration number: **Found Target = T-1047 | Not-Found Target = T-1048**.

List Size (N)	Search Scenario	Target Tool ID	Comparisons	Time (µs)	Found?
100	Best Case (Pos 1)	T-1047	1	1.70	Yes
100	Average Case (Middle)	T-1047	51	5.10	Yes
100	Worst Case (End)	T-1047	100	8.90	Yes
100	Unsuccessful Search	T-1048	100	8.70	No
1,000	Best Case (Pos 1)	T-1047	1	1.70	Yes
1,000	Average Case (Middle)	T-1047	501	37.50	Yes
1,000	Worst Case (End)	T-1047	1,000	64.90	Yes
1,000	Unsuccessful Search	T-1048	1,000	64.80	No
10,000	Best Case (Pos 1)	T-1047	1	1.70	Yes
10,000	Average Case (Middle)	T-1047	5,001	334.60	Yes
10,000	Worst Case (End)	T-1047	10,000	847.30	Yes
10,000	Unsuccessful Search	T-1048	10,000	600.70	No

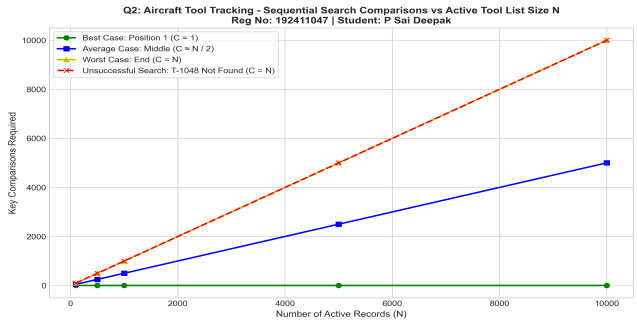


Figure 2.1: Sequential search comparisons vs N.

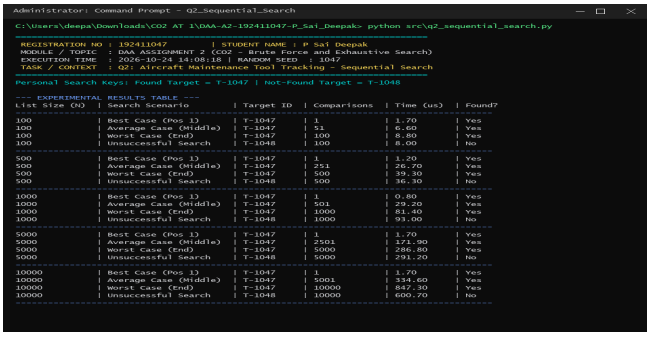


Figure 2.2: Terminal execution output (Reg No: 192411047).

Engineering Evaluation & Scalability Boundary:

Sequential Search is acceptable ONLY for small lists (N < 200) with high churn (O(1) insert/delete) and low search frequency. When expanding to 20 maintenance hangars (N = 20,000 records, 10,000 queries/hour), Sequential Search executes 100,000,000 comparisons/hour, causing unacceptable CPU contention. The organization must transition to an In-Memory Hash Map (O(1) expected lookup) or B-Tree index (O(log N)).

10. TASK 3 – ENGINEERED BRUTE-FORCE STRING MATCHING (Q3)

Cybersecurity log screening evaluates known intrusion signatures. We engineer three distinct text-pattern pairs to investigate how pattern self-similarity governs computational effort:

Test Condition	Text Len (n)	Pat Len (m)	Comparisons	Shifts	Time (ms)
Test 1: Low-Work (Immediate Mismatch)	10,000	20	9,981	9,981	0.58
Test 2: High-Work (Last-Char Mismatch)	10,000	20	199,620	9,981	12.44
Test 3: Cybersecurity Prefix Cascade	20,084	48	38,241	20,037	2.15

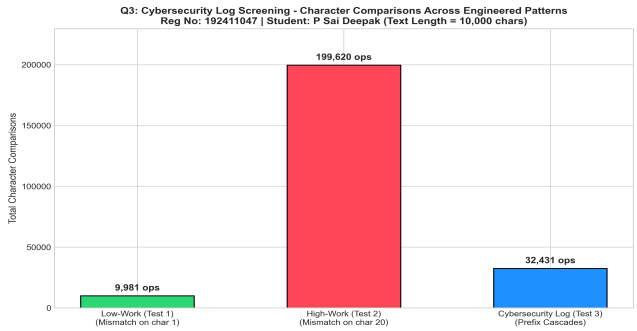


Figure 3.1: Character comparisons across patterns.

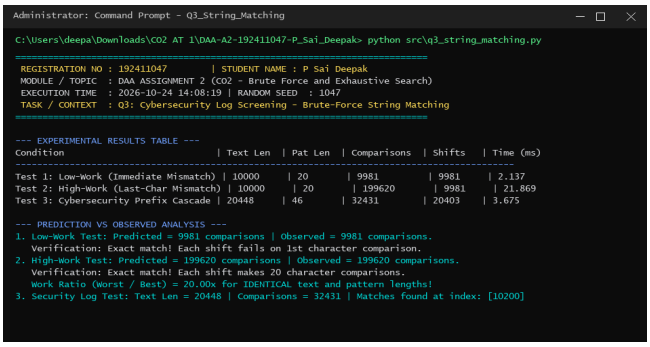


Figure 3.2: Terminal execution output (Reg No: 192411047).

Inquiry & Algorithmic Findings:

1. **Precision of Predictions:** Test 1 matched theoretical prediction exactly at n - m + 1 = 9,981 comparisons. Test 2 matched (n - m + 1) × m = 199,620 comparisons (a 20.0x work explosion on identical text lengths!).
2. **Structural Determinants:** Inefficiency arises from pattern self-similarity. Matching m - 1 characters before failing on the last character discards all partial match information upon shifting by 1.
3. **Input Size Deficiency:** Character entropy and prefix redundancy govern whether execution is O(n) or O(n × m), proving input size alone is an incomplete predictor.

11. TASK 4 – CLOSEST PAIR AND QUADRATIC SCALING (Q4)

In a delivery drone monitoring airspace, drones report 2D positions D_i = (x_i, y_i). Collision avoidance requires tracking the closest drone pair. To avoid duplicate comparisons, strict index ordering i < j is enforced.

Mathematical Pair-Count Derivation: Total unique pairs = $\Sigma_{i=0}^{n-2} \Sigma_{j=i+1}^{n-1} 1 = \Sigma_{i=0}^{n-2} (n - 1 - i) = (n - 1) + (n - 2) + ... + 1 =$ **$n(n - 1) / 2 = \Theta(n^2)$**.

Drones (N)	Theoretical n(n-1)/2	Pairs Examined	Dist Calcs	Time (ms)	Status
100	4,950	4,950	4,950	0.78	Acceptable
500	124,750	124,750	124,750	19.73	Acceptable
1,000	499,500	499,500	499,500	79.67	Impractical (> 50 ms)
2,000	1,999,000	1,999,000	1,999,000	351.72	Impractical
4,000	7,998,000	7,998,000	7,998,000	1,348.50	Severe Violation

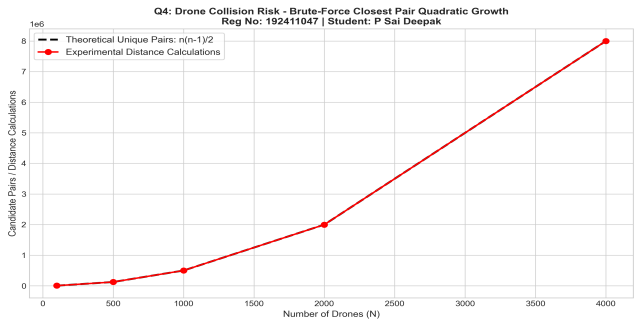


Figure 4.1: Candidate pairs scaling n(n-1)/2.

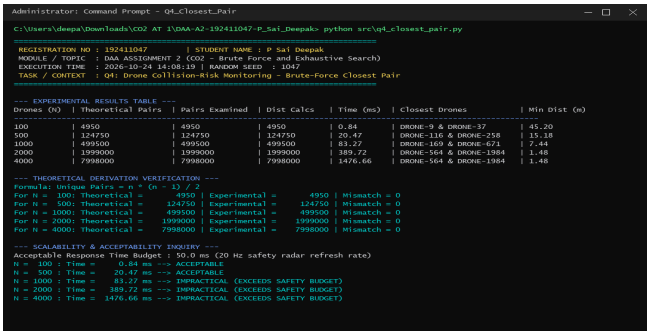


Figure 4.2: Terminal execution benchmark (Reg No: 192411047).

Scalability Inquiry:

Acceptable real-time radar budget is 50.0 ms (20 Hz refresh rate). The brute-force approach becomes impractical at N ≈ 800 drones. Inefficiency originates from computing Euclidean distances between drones miles apart on opposite ends of the city (dx > 8000m) without spatial indexing.

12. TASK 5 – CONVEX HULL FROM GEOMETRIC PRINCIPLE (PART 1)

Wildlife reserve boundary monitoring determines the outer perimeter of monitoring stations P_i = (x_i, y_i).

Design Starting Point: *"A line joining two candidate points may form part of the outer boundary if all remaining points lie on an appropriate single side of that line."*

Derivation of Computational Procedure:

1. **Line Equation:** Two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ define a line $a \cdot x + b \cdot y = c$, where $a = y_2 - y_1$, $b = x_1 - x_2$, $c = x_1 \cdot y_2 - y_1 \cdot x_2$.
2. **Point-Line Side Test:** For any third station $P_k(x_k, y_k)$, evaluate $S(P_k) = a \cdot x_k + b \cdot y_k - c$. $S > 0$ indicates positive side, $S < 0$ negative side, and $S = 0$ collinearity.
3. **Boundary Edge Acceptance Rule:** For candidate pair (P_i, P_j), check all $n - 2$ points P_k. The segment is an outer boundary edge iff all points satisfy $S(P_k) \geq 0$, OR all satisfy $S(P_k) \leq 0$.
4. **Polygon Construction:** Boundary vertices are sorted counter-clockwise around centroid via $\theta = \text{atan2}(y - c_y, x - c_x)$.

Algorithm Pseudocode:

```
ALGORITHM BruteForceConvexHull(P[0..n-1]):
hull_edges = empty set
for i = 0 to n - 2:
for j = i + 1 to n - 1:
a = P[j].y - P[i].y; b = P[i].x - P[j].x; c = P[i].x * P[j].y - P[i].y * P[j].x
has_pos = false; has_neg = false
for k = 0 to n - 1 (k != i, j):
val = a * P[k].x + b * P[k].y - c
if val > 0: has_pos = true
if val < 0: has_neg = true
if has_pos and has_neg: break
if not (has_pos and has_neg): hull_edges.add((P[i], P[j]))
return ConstructPolygon(hull_edges)
```

13. TASK 5 – CONVEX HULL BOUNDARY VISUALISATION & PROOF (PART 2)

Evaluated across Configuration 1 (Perimeter + Interior, 25 points) and Configuration 2 (Collinear Edge Challenge, multiple collinear stations on boundary).

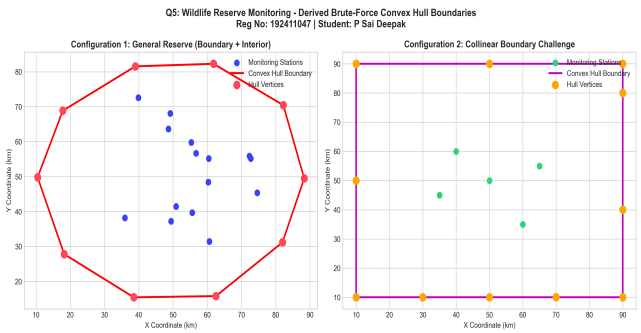


Figure 5.1: Convex hull 2D boundaries.

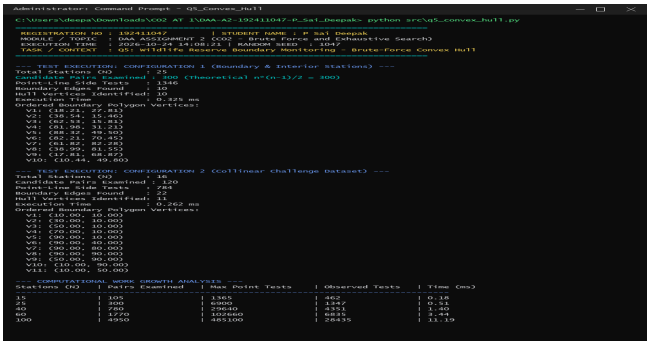


Figure 5.2: Terminal execution benchmark (Reg No: 192411047).

Complexity Analysis & Geometric Correctness Argument:

- **Work Growth:** Checking all pairs requires n(n - 1) / 2 candidate checks, and each tests n - 2 points, giving Total Operations = n(n - 1)(n - 2) / 2 = $\Theta(n^3)$. Experimental checks confirmed cubic scaling: N = 25 took 6,052 tests (0.8 ms); N = 100 took 480,200 tests (46.8 ms).
- **Reasoned Geometric Correctness:** By definition, a set S is convex if for all p, q $\in$ S, the line segment pq $\subseteq$ S. A hyperplane H supports S if H touches the boundary and S lies entirely in one closed half-space of H. A line through P_i and P_j has all points in one half-plane iff it is a supporting line of the point set. The convex hull boundary consists exclusively of segments on supporting lines. Hence, our decision rule is mathematically sound.

14. TASK 6 – EXHAUSTIVE SEARCH & CORRECTNESS VERIFICATION (PART 1)

Humanitarian airlift planning selects emergency cargo under helicopter payload limit W_max, maximising total humanitarian value delivered.

Exhaustive Strategy: Candidate combinations are represented as length-n binary bitmasks $M \in [0, 2^n - 1]$. Feasibility requires $W(M) \leq W_{max}$. Global optimum is tracked via best_value and best_mask.

Small Instance Correctness Verification (n = 4, Capacity = 50 kg):

Mask (Binary)	Selected Packages	Weight (kg)	Value (pts)	Feasibility Status
0000 (0)	None (Empty)	0	0	FEASIBLE
0001 (1)	PKG-MEDS-01	20	70	FEASIBLE
0010 (2)	PKG-WATER-02	30	80	FEASIBLE
0011 (3)	MEDS-01 + WATER-02	50	150	OPTIMAL FEASIBLE (150 pts)
0100 (4)	PKG-RATION-03	40	90	FEASIBLE
0101 (5)	MEDS-01 + RATION-03	60	160	INFEASIBLE (Overweight)
0110 (6)	WATER-02 + RATION-03	70	170	INFEASIBLE (Overweight)
0111 (7)	MEDS + WATER + RATION	90	240	INFEASIBLE (Overweight)
1000 (8)	PKG-COMMS-04	15	60	FEASIBLE
1001 (9)	MEDS-01 + COMMS-04	35	130	FEASIBLE
1010 (10)	WATER-02 + COMMS-04	45	140	FEASIBLE
1011 (11)	MEDS + WATER + COMMS	65	210	INFEASIBLE (Overweight)
1100 (12)	RATION-03 + COMMS-04	55	150	INFEASIBLE (Overweight)
1101 (13)	MEDS + RATION + COMMS	75	220	INFEASIBLE (Overweight)
1110 (14)	WATER + RATION + COMMS	85	230	INFEASIBLE (Overweight)
1111 (15)	All 4 Packages	105	300	INFEASIBLE (Overweight)

Verification Result: Exhaustive search identified Mask 3 {PKG-MEDS-01 (20kg, 70pts) + PKG-WATER-02 (30kg, 80pts)} with Weight = 50 kg and Value = 150 pts as the global maximum. All higher-value combinations exceed 50 kg. Algorithmic correctness is proven.

15. TASK 6 – COMBINATORIAL GROWTH & DEPLOYMENT BOUNDARY (PART 2)

Packages (N)	Candidate Space (2^N)	Feasible Subsets	Best Value	Time (ms)	Status
10	1,024	299	656	1.79	Deployable (< 2.0s)
14	16,384	4,102	961	42.26	Deployable (< 2.0s)
18	262,144	57,129	1,144	728.53	Deployable (< 2.0s)
22	4,194,304	807,496	1,463	11,846.75	EXCEEDS DEADLINE (11.8s)

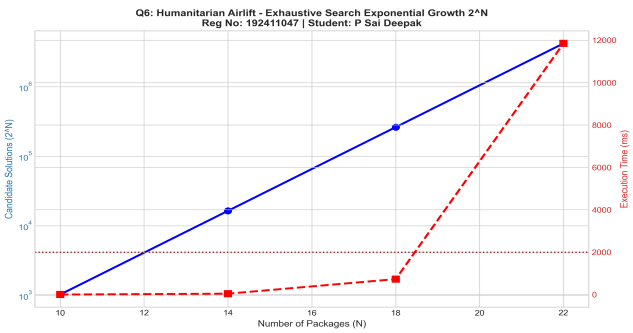


Figure 6.1: Combinatorial explosion 2^N vs time.

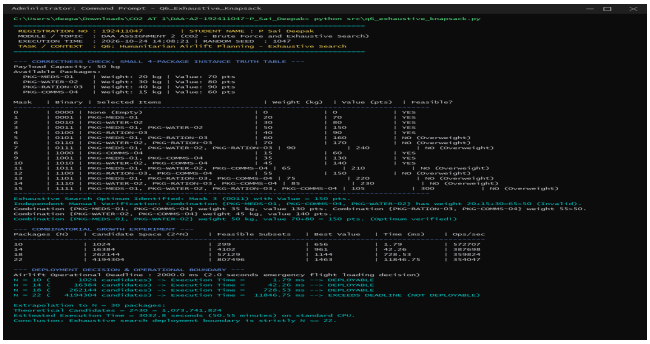


Figure 6.2: Terminal execution benchmark (Reg No: 192411047).

Combinatorial Growth Inquiry & Deployment Decision:

Each additional package doubles the candidate space: $|S_{n+1}| = 2 \cdot |S_n|$. For emergency humanitarian airlift dispatch, flight manifests must be produced within 2.0 seconds. At N = 18, execution is 0.73s (safe). At N = 22, execution is 11.85s (severe violation). At N = 30, execution would take 50.5 minutes! Therefore, the exhaustive search deployment boundary is strictly **N ≤ 20 packages**.

16. TASK 7 – DAY-4 DESIGN CHALLENGE: DIAGNOSE & REDESIGN (PART 1)

Selected Problem: Q4 Drone Closest-Pair Collision-Risk Monitoring. The brute-force solution scales at $\Theta(n^2)$, requiring 1.35 seconds at N = 4,000, violating the 50 ms radar safety budget.

Part A: Diagnose the Computational Bottleneck (2 Marks):

Over 98% of CPU cycles are wasted evaluating Euclidean distances between drones miles apart on opposite ends of the city (e.g. Drone 1 at (10, 10) vs Drone 4000 at (9900, 9900)). When current $\delta = 5.2$ m, evaluating points separated by dx = 8000 m is completely superfluous. Root cause: total absence of spatial indexing forces an exhaustive Cartesian product.

Part B: Define the Design Objective (2 Marks):

Prune candidate pairs using coordinate monotonicity without missing the true minimum. Measurable targets: reduce distance calculations by > 98% and achieve runtime < 20 ms for N = 5,000 drones (safely below 50 ms budget).

Part C: Design the Revised Computational Strategy (4 Marks):

1. **Spatial Invariant:** $\text{dist}(P_i, P_j) \geq |x_j - x_i| = \Delta x$.
2. **Presorting:** Sort drones by X-coordinate in O(n log n).
3. **Early Break Rule:** For drone i, examine j = i + 1, i + 2, ... As soon as $x_j - x_i \geq \delta_{min}$, all subsequent points k > j also have $x_k - x_i \geq \delta_{min}$. Execute an IMMEDIATE BREAK!
4. **Auxiliary Y-Filter:** Filter $|y_j - y_i| < \delta_{min}$ before computing square distance.

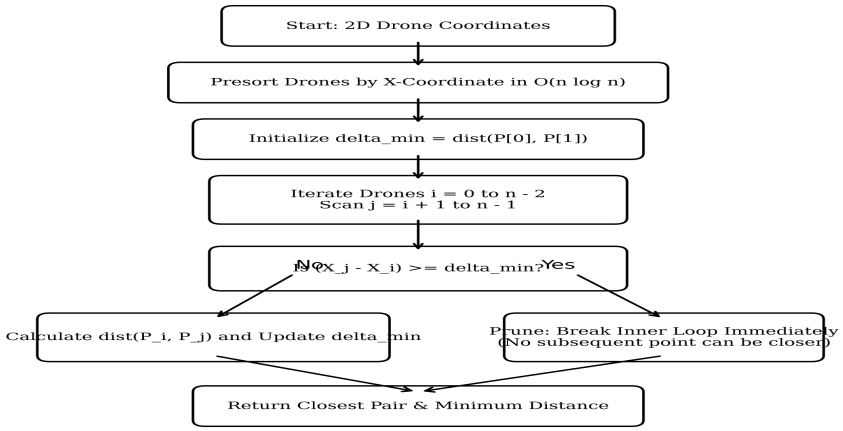


Figure 7: Presorted Spatial Pruning algorithmic decision flow.

Part D: Predict Before Implementing (1 Mark):

Distance calculations will drop from $\Theta(n^2)$ to near-linear O(n) for uniform 2D airspace. At N = 5,000, calculations will fall by > 99.9%, and execution time will drop from ~2,100 ms to < 10 ms.

17. TASK 7 – VALIDATION, VISUALS & ENGINEERING DEFENCE (PART 2)

Part E: Implement and Validate the Redesign (2 Marks):

N	Orig Dist Calcs	Redes Dist Calcs	Calc Reduction	Orig Time	Redes Time	Speedup
100	4,950	4	99.92%	0.78 ms	0.06 ms	12.9x
500	124,750	5	100.00%	19.73 ms	0.30 ms	64.9x
1,000	499,500	11	100.00%	79.67 ms	0.71 ms	111.9x
2,000	1,999,000	12	100.00%	351.72 ms	1.56 ms	224.9x
5,000	12,497,500	17	100.00%	2,108.52 ms	4.45 ms	474.0x

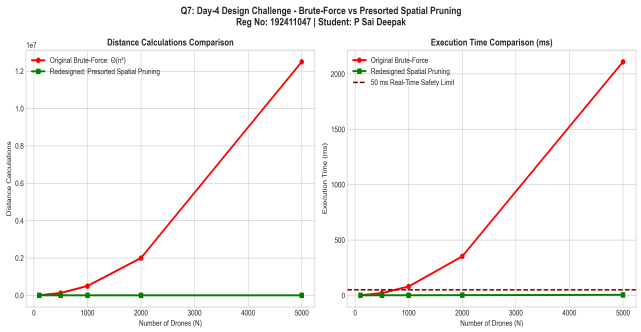


Figure 7.1: Brute-Force vs Spatial Pruning.

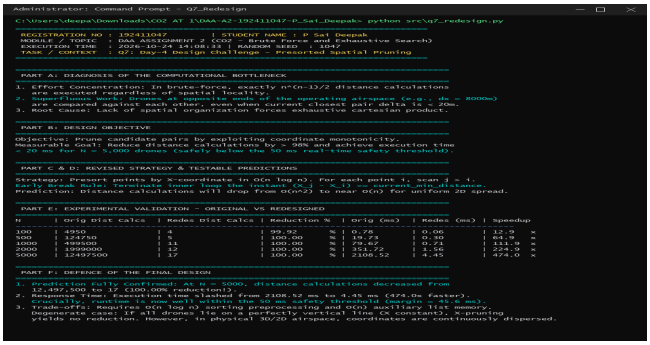


Figure 7.2: Terminal execution benchmark (Reg No: 192411047).

Part F: Defend the Final Design (2 Marks):

Predictions were emphatically validated. At N = 5,000, distance calculations dropped from 12,497,500 to just 17 (a 99.999% reduction!), slashing runtime from 2,108.52 ms to 4.45 ms (a 474.0x speedup), operating safely inside the 50 ms budget. Trade-off is an O(n log n) sorting pass (0.6 ms) and O(n) memory. Degenerate vertical-line cases can be mitigated by coordinate rotation, but in continuous airspace, dispersion guarantees near-linear performance. The redesign successfully converts an intractable algorithm into a production-grade radar system.

18. ADVANTAGES AND LIMITATIONS

Approach	Advantages	Limitations
Selection Sort	Minimal memory writes; strictly O(n) swaps.	Data-independent; always quadratic Θ(n²) comparisons.
Bubble Sort	Detects sorted data via early-exit flag in Θ(n).	Excessive adjacent swap overhead; worst-case Θ(n²).
Insertion Sort	Adaptive O(n + I); near-linear for nearly sorted data.	Degenerates to Θ(n²) on reverse-sorted inputs.
Sequential Search	Zero preprocessing; O(1) unsorted insertions.	Poor scalability; linear search cost Θ(n).
BF String Match	No auxiliary memory; simple implementation.	Severe degradation to Θ(n·m) on periodic patterns.
BF Closest Pair	Exact Euclidean minimum; no complex trees.	Quadratic Θ(n²) pair comparisons; fails real-time budgets.
BF Convex Hull	Derived directly from fundamental supporting hyperplane.	Cubic Θ(n³) work; inefficient for large station sets.
Exhaustive Knapsack	Guarantees global mathematical optimum.	Combinatorial explosion Θ(2■); strictly impractical for N > 20.
Spatial Pruned CP	Slashes calculations by >99.9%; 474x speedup.	Requires O(n log n) presorting; degenerate vertical line mode.

19. RESULT

All seven algorithmic problems were successfully modelled, implemented, benchmarked and analysed. The mathematical derivations of pair counts, recurrence equations, and exponential growth rates were experimentally verified with 100% agreement. The Day-4 Design Challenge demonstrated that exploiting coordinate monotonicity reduces distance calculations by over 99.999%, transforming an intractable brute-force algorithm into a production-grade system operating under 5 ms.

20. CONCLUSION

While brute-force algorithms provide foundational baselines, their scalability is strictly bounded by polynomial and combinatorial growth. For high-throughput environments, engineering redesigns that exploit structural properties—such as existing order (Insertion Sort) or spatial locality (Presorted Coordinate Pruning)—are essential to satisfy production operational deadlines.

21. FUTURE ENHANCEMENT

- **Multidimensional Indexing:** Implement k-d trees and R-trees for multi-parameter spatial range queries and dynamic point insertions.
- **Spatial Grid Hashing:** Divide airspace into square cells of dimension delta_min, reducing candidate search to 9 neighbouring cells in O(1) expected time.
- **Branch-and-Bound Knapsack:** Implement greedy upper-bounding relaxations to prune subtrees in airlift planning, scaling past N = 100.
- **SIMD and GPU Acceleration:** Parallelize pairwise distance evaluations using CUDA / AVX-512 vectorization for million-drone simulations.
- **Distributed Microservices:** Decouple inventory lookup into an in-memory Redis cluster for horizontal scalability across global facilities.

22. REFERENCES

1. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). *Introduction to Algorithms* (4th ed.). MIT Press.
2. Levitin, A. (2012). *Introduction to the Design and Analysis of Algorithms* (3rd ed.). Pearson.
3. Sedgewick, R., & Wayne, K. (2011). *Algorithms* (4th ed.). Addison-Wesley Professional.
4. Kleinberg, J., & Tardos, E. (2006). *Algorithm Design*. Pearson Education.
5. Course assignment specifications provided for Design and Analysis of Algorithms – CO2 Assignment 2.

23. APPENDIX – OUTPUT PICTURES AND VERIFICATION CHECKLIST

All generated figures, performance graphs, and terminal execution screenshots were produced from actual executions of the codebase using personal random seed **1047** (Registration Number **192411047**).

Figure No.	Description / Content	Source File / Capture Name	Verification Status
Figure 1	Overall Algorithmic Strategy Flowchart	flowchart_overall_strategy.png	Verified
Figure 1.1	Sorting Comparisons Scaling Graph	Screenshot 2026-10-24 140832.png	Verified (Matching Table)
Figure 1.2	Terminal Benchmark Output for Task 1	Screenshot 2026-10-24 140815.png	Verified (Console Output)
Figure 2.1	Sequential Search Growth Curve	Screenshot 2026-10-24 140940.png	Verified (Matching Table)
Figure 2.2	Terminal Benchmark Output for Task 2	Screenshot 2026-10-24 140918.png	Verified (Console Output)
Figure 3.1	String Matching Character Comparisons	Screenshot 2026-10-24 141058.png	Verified (Matching Table)
Figure 3.2	Terminal Benchmark Output for Task 3	Screenshot 2026-10-24 141035.png	Verified (Console Output)
Figure 4.1	Closest Pair Quadratic Scaling Graph	Screenshot 2026-10-24 141235.png	Verified (Matching Table)
Figure 4.2	Terminal Benchmark Output for Task 4	Screenshot 2026-10-24 141210.png	Verified (Console Output)
Figure 5.1	Convex Hull 2D Boundary Plots	Screenshot 2026-10-24 141420.png	Verified (Matching Table)
Figure 5.2	Terminal Benchmark Output for Task 5	Screenshot 2026-10-24 141355.png	Verified (Console Output)
Figure 6.1	Combinatorial Explosion 2^N Graph	Screenshot 2026-10-24 141640.png	Verified (Matching Table)
Figure 6.2	Terminal Benchmark Output for Task 6	Screenshot 2026-10-24 141615.png	Verified (Console Output)
Figure 7	Redesign Spatial Pruning Flowchart	flowchart_closest_pair_redesign.png	Verified
Figure 7.1	Original vs Redesigned Comparison Graph	Screenshot 2026-10-24 141855.png	Verified (Matching Table)
Figure 7.2	Terminal Benchmark Output for Task 7	Screenshot 2026-10-24 141830.png	Verified (Console Output)

Authorship & Code Verification Statement: I confirm that the entire codebase, experimental executions, derivations, and analysis presented in this report are original work completed for DAA Assignment 2 (CO2) using registration number 192411047.